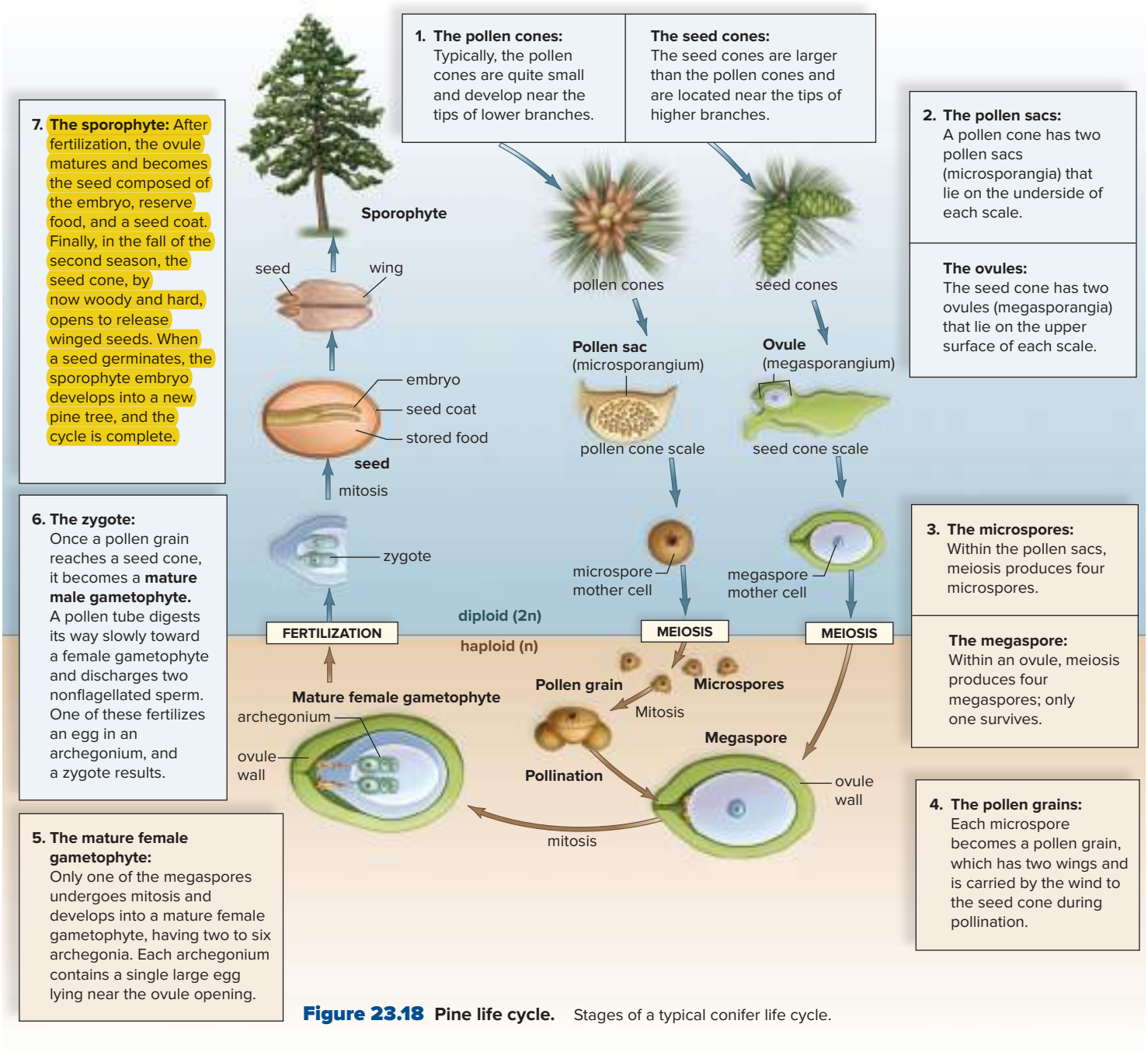


Figure 23.18 Pine life cycle. Stages of a typical conifer life cycle.

recessed stomata. Note that in the life cycle of the pine (Fig. 23.18) the sporophyte is dominant, pollen grains are windblown, and the seed is what is dispersed. Conifers are **monoecious,** which means that a single plant carries both male and female reproductive structures.

Pines, in particular, are well-known for their beauty and pleasant smell; they make attractive additions to parks and gardens. Pines have medicinal value in that the needles and bark are rich in vitamins A and C and can be used to create teas that ease symptoms of a cold or cough. Large pine seeds, called pine nuts, are sometimes harvested for use in cooking, and pine oil is used to scent a number of household and personal care products, such as room sprays and deodorant.

Cycads

Cycads include ten genera and 300 species of distinctive gymnosperms. The cycads are native to tropical and subtropical forests. *Zamia pumila,* found in Florida, is the only species of cycad native to North America. Cycads are commonly used in landscaping. One species, *Cycas revoluta,* referred to as the sago palm, is a common landscaping plant. Their large, finely divided leaves grow in clusters at the top of the stem, and therefore they resemble palms or ferns, depending on their height. The trunk of a cycad is unbranched, even if it reaches a height of 15–18 m, as is possible in some species.

Unlike conifers, cycads have pollen and seed cones on separate plants. The cones, which grow at the top of the stem surrounded by the leaves, can be huge—more than a meter long with a weight of

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